ASSOCIATING MULTI-SCALE RECEPTIVE FIELDS FOR FINE-GRAINED RECOGNITION

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1. Background
2. Cross-layer Non-local Module
3. Experiment
4. Conclusion

1 Background
Fine-Grained Recognition
Existing Works

2 Cross-layer Non-local Module

3 Experiment

4 Conclusion

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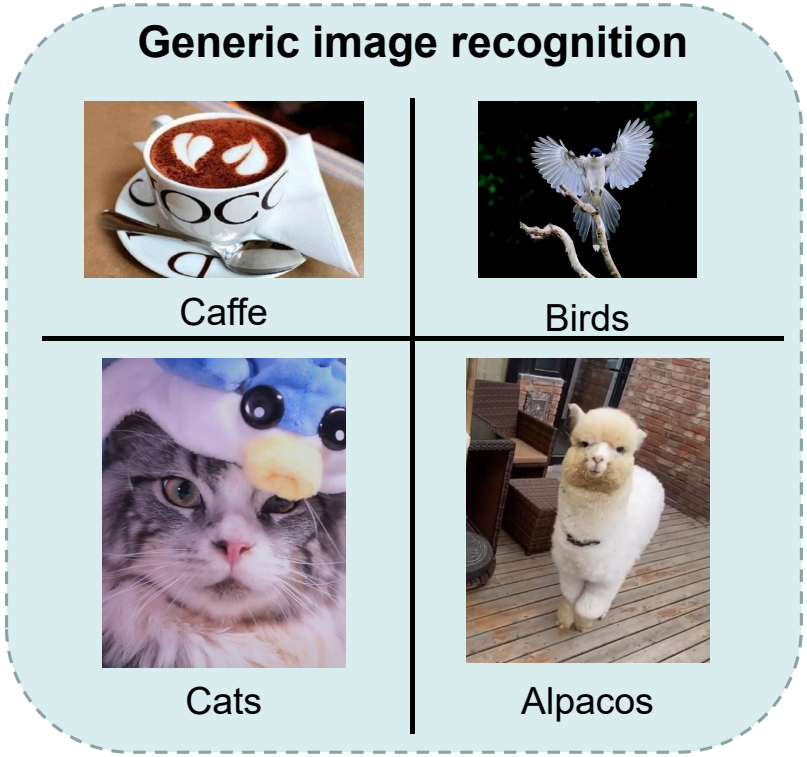

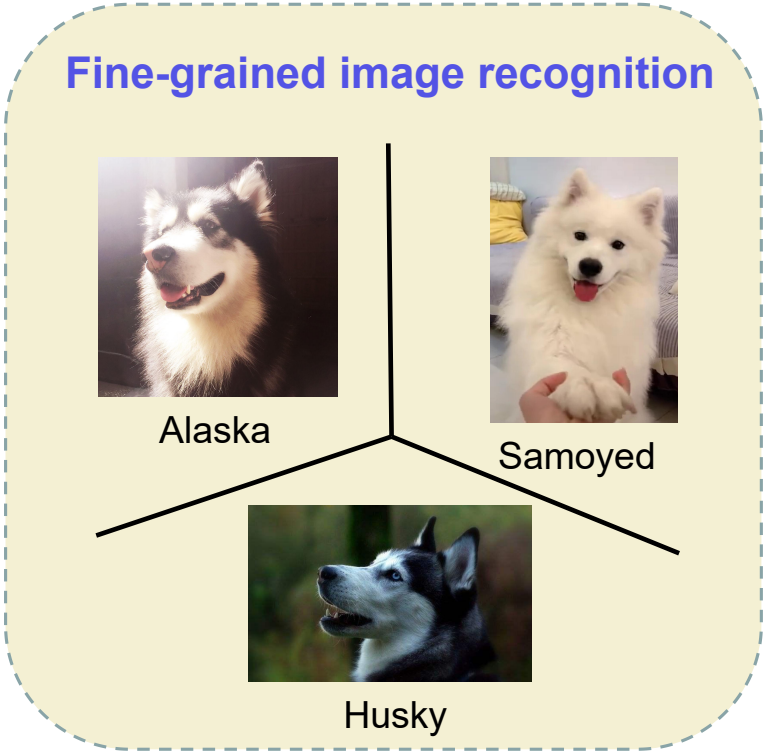

Generic Image Recognition:
coarse-grained meta-categories
visually quite different

Fine-grained Image Recognition:
sub-categories of one meta-category
visually quite similar

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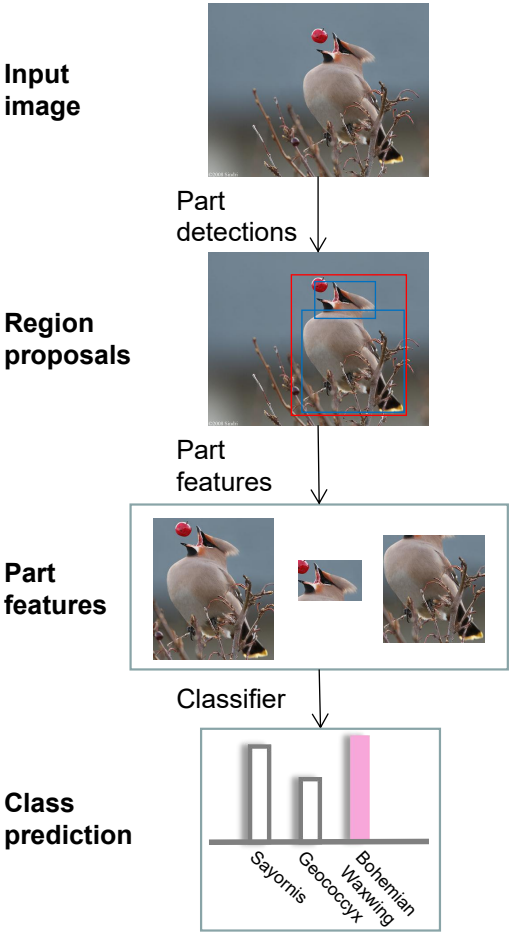
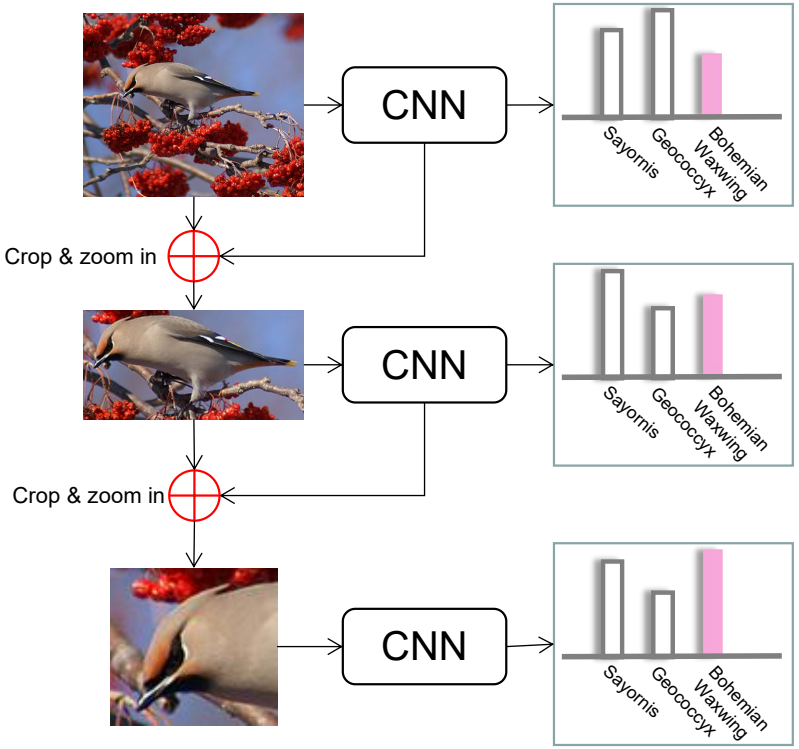
Existing Works



(a) Localization-based method
Part-based R-CNN



(b) Attention-based method
RACNN

Extracting and fusing part features have become the key of fined-grained image recognition.

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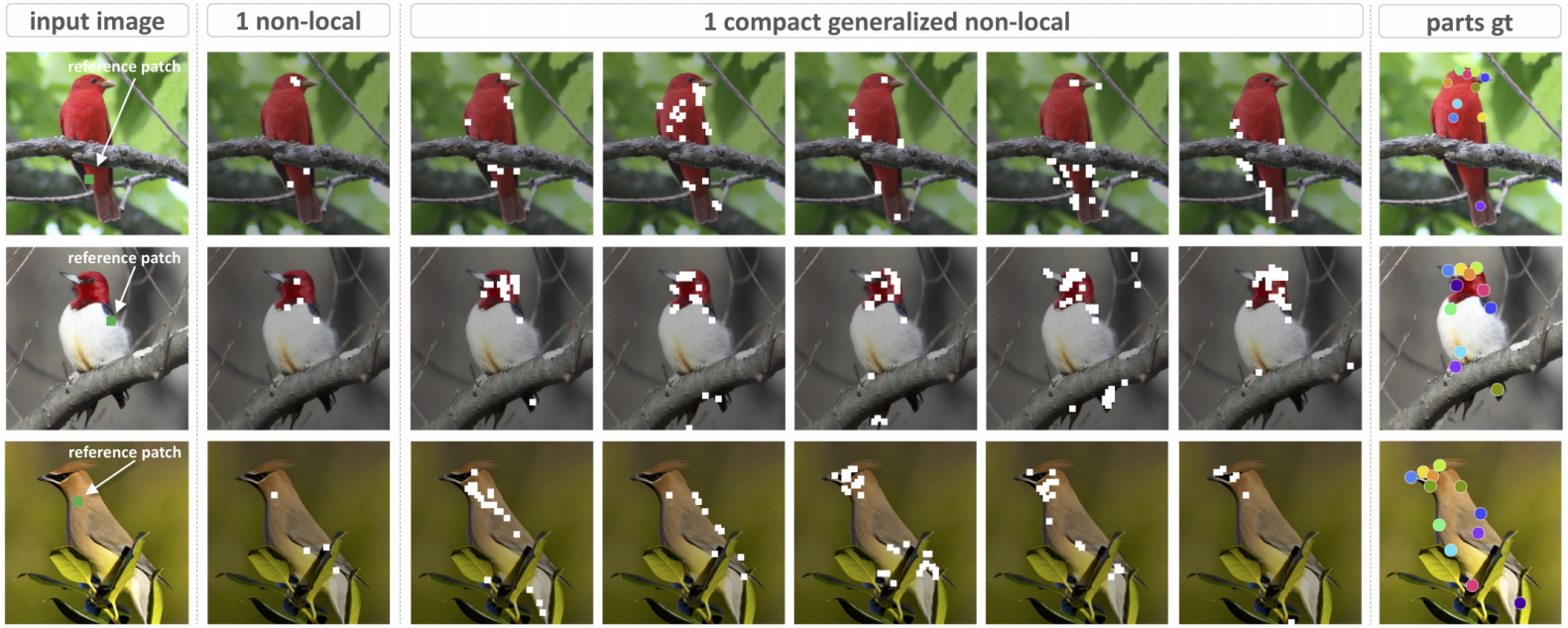
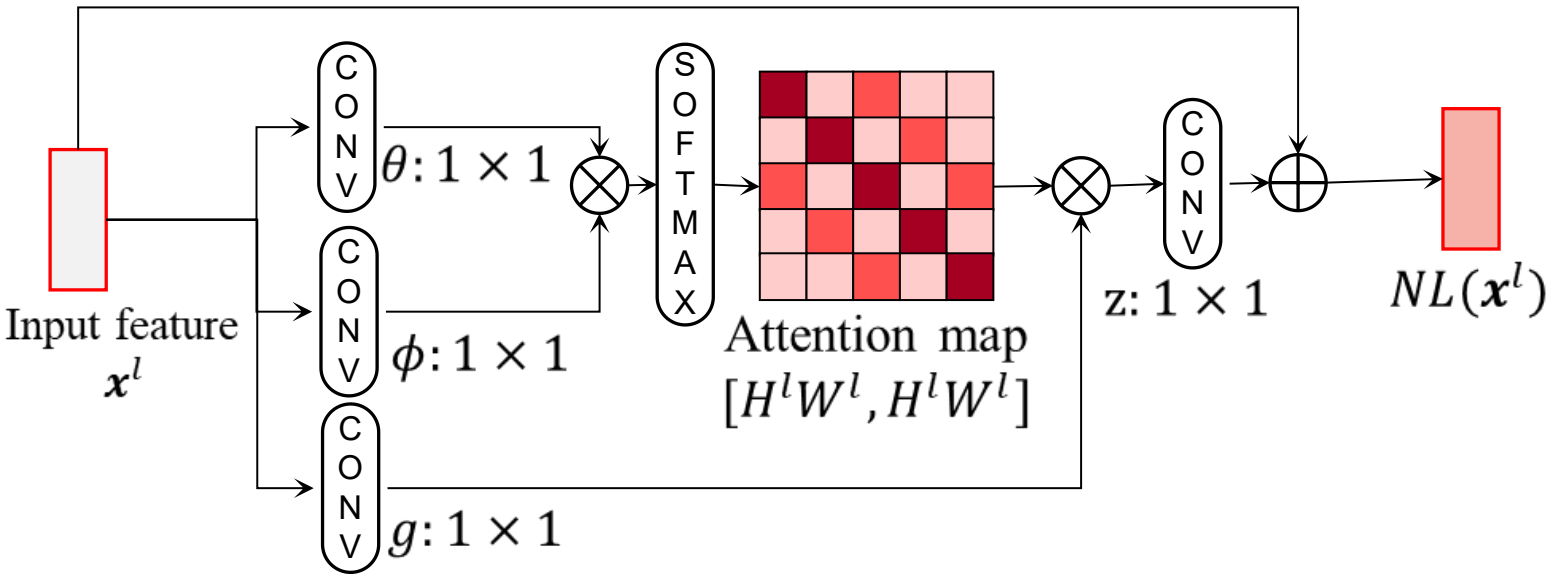


Non-local is useful not only in Generic Image Recognition, but in Fine-Grained Image Recognition.

[1] Kaiyu Yue, Ming Sun, Yuchen Yuan, Feng Zhou, Errui Ding, and Fuxin Xu, "Compact generalized non-local network," in NeurIPS, 2018.



Formulation:

$$NL(\mathbf{X}^l) = \mathbf{X}^l + z(f(\theta(\mathbf{X}^l), \phi(\mathbf{X}^l))g(\mathbf{X}^l)). \quad (1)$$

$\mathbf{X}^l \in \mathbf{R}^{H^l W^l \times C^l}$: inputed feature map;
$\theta(\cdot)$, $\phi(\cdot)$, $g(\cdot)$ and $z(\cdot)$: learnable transformations, general 1×1 convolution;
$f(\cdot, \cdot)$: a pairwise function computes the attention map.

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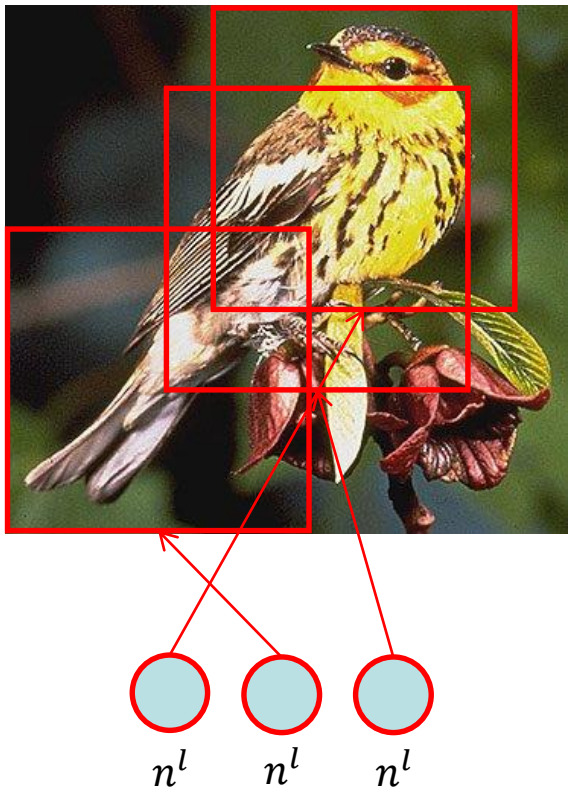
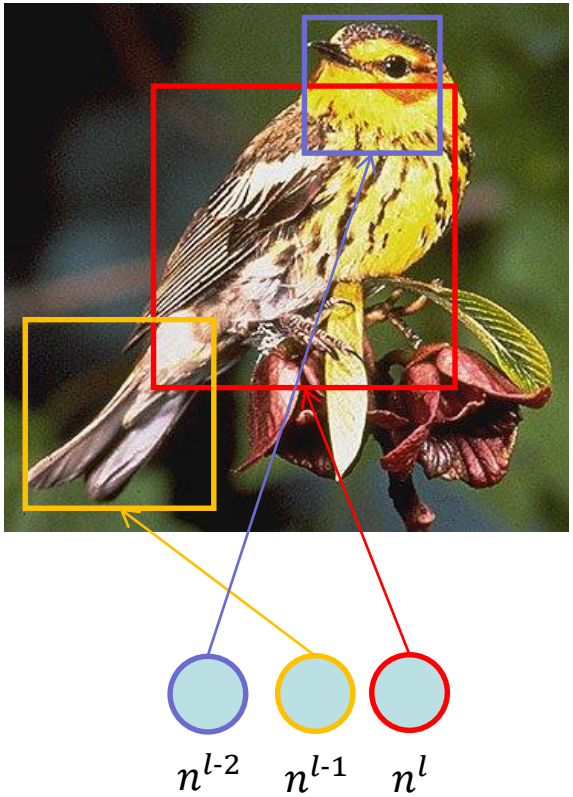


(a) Receptive fields in the same layer



(b) Receptive fields in different layers

Receptive field: the region of input layer determines the output result of a neuron.

n^l: the neurons in l-th layer.

Over-sized receptive fields $\rightarrow$ more background noises.

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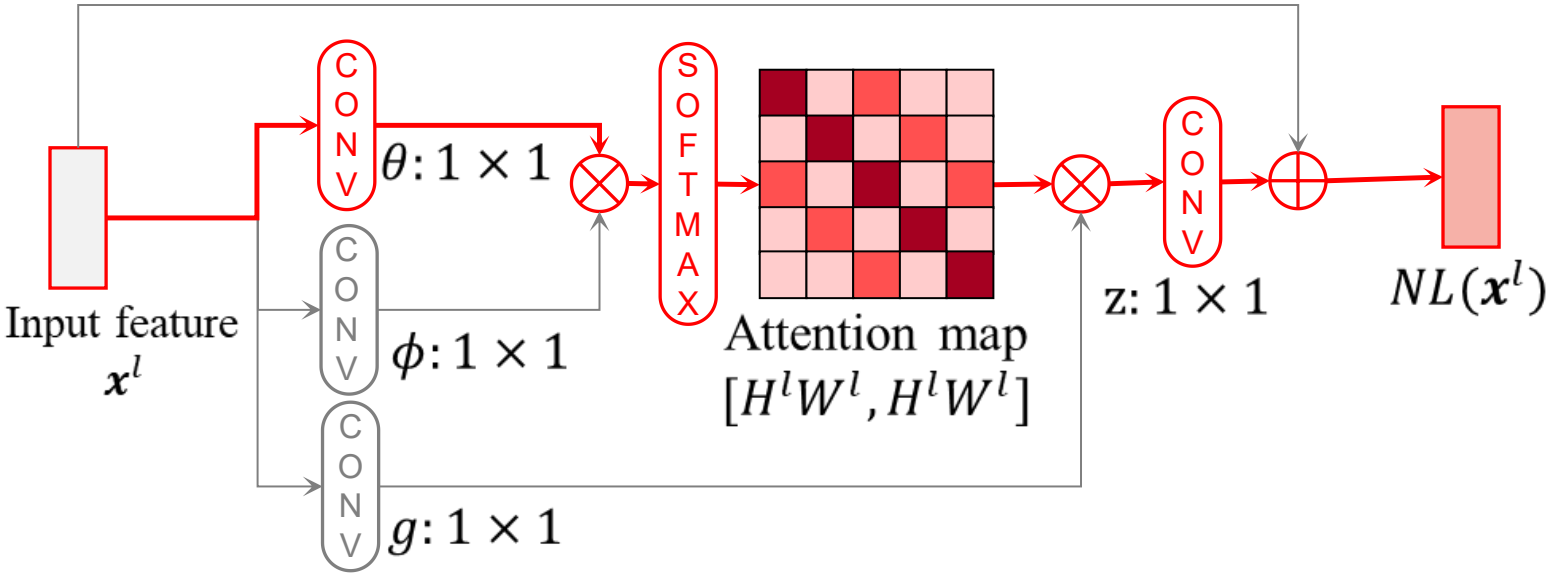
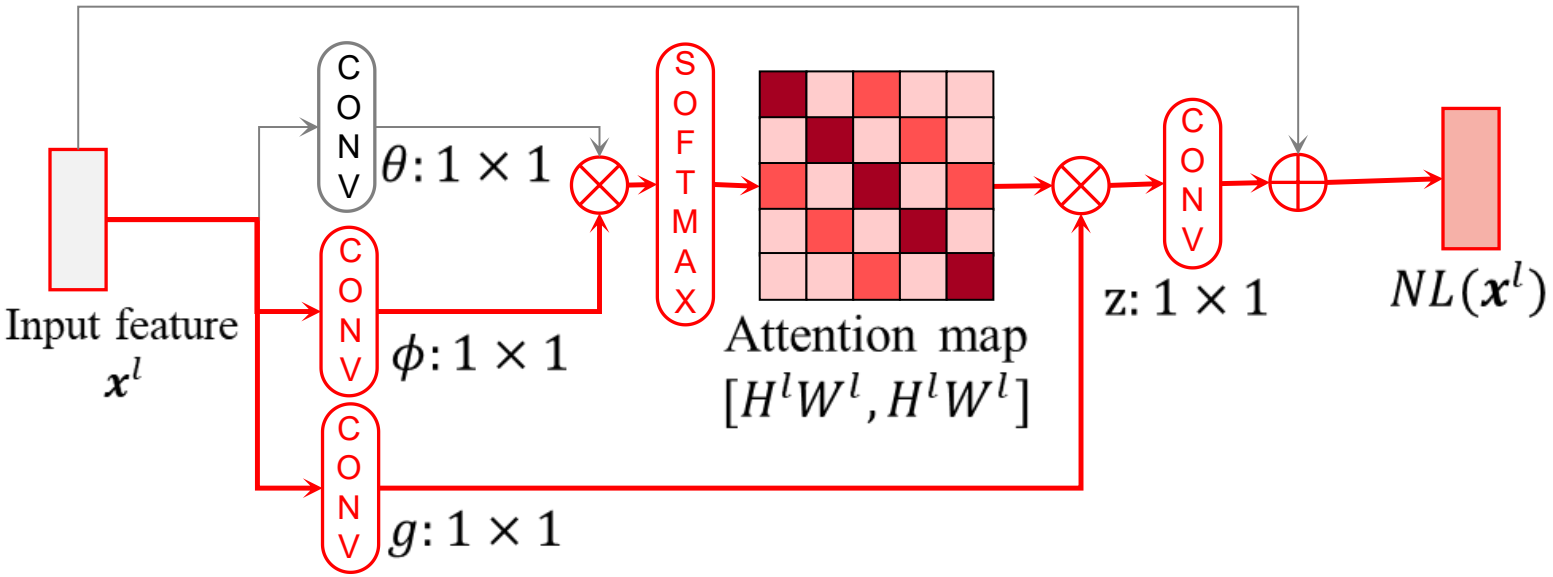
Two data flows of NL:



One flow:$\mathbf{X}^q \in \mathbf{R}^{H^q W^q \times C^q}$ is only inputted to θ convolution in order to compute the attention map.

Two data flows of NL:



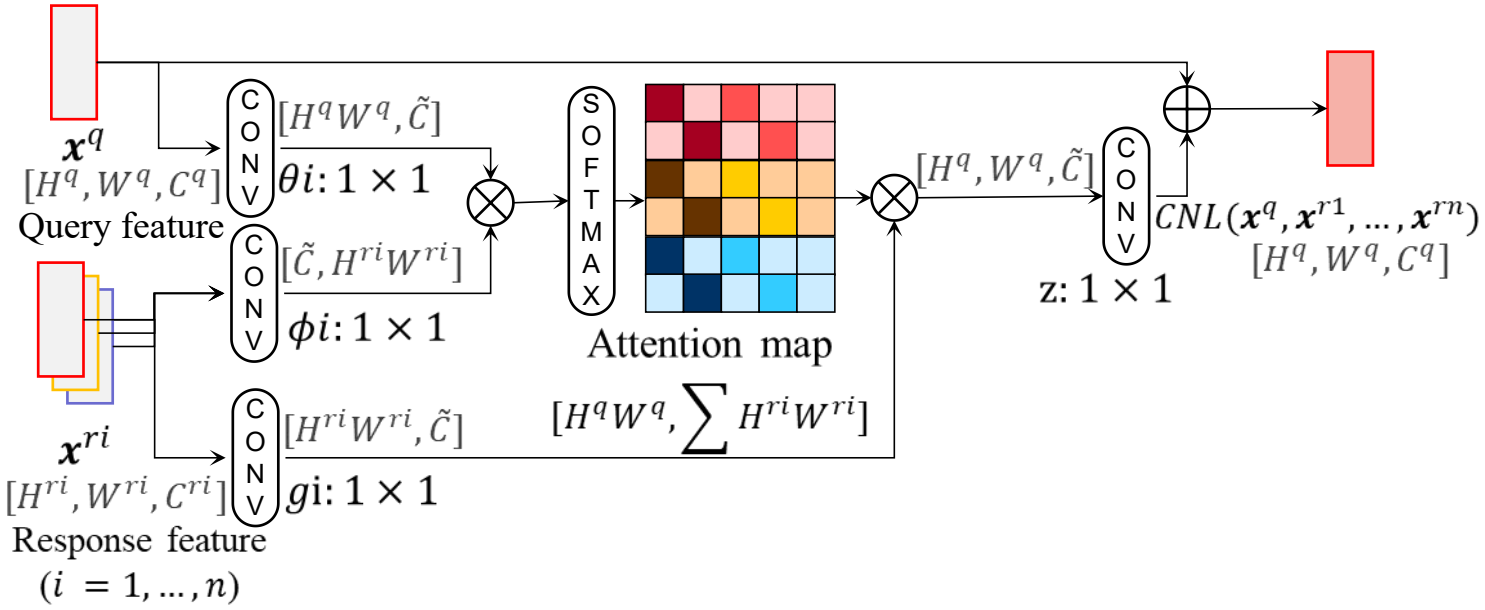
The other flow: $\mathbf{X}^r \in \mathbf{R}^{H^r W^r \times C^r}$ is not only inputted to $\phi(\cdot)$ for the attention map, but also is inputted to $g(\cdot)$ for the response results.

More general, two-variables form of NL:

$$NL_{bi}(\mathbf{X}^q, \mathbf{X}^r) = \mathbf{X}^l + z(f(\theta(\mathbf{X}^q), \phi(\mathbf{X}^r))g(\mathbf{X}^r)). \tag{2}$$

Considerring One deep layer as query layer, and associating it with several shadow response layers.

Cross-layer Non-local Module



$$CNL(\mathbf{X}^q, \mathbf{X}^{r1}, \cdots, \mathbf{X}^{rn}) = \mathbf{X}^q + \sum \mathbf{Z}^{ri}, \tag{3}$$

where $\mathbf{Z}^{ri} = z(f(\theta(\mathbf{X}^q), \phi i(\mathbf{X}^{ri}))gi(\mathbf{X}^{ri}))$.

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Method	top1	top5	#param($\times 10^7$)	Flops($\times 10^9$)
ResNet50	84.05	96.00	2.39	1.64
+5NL	85.10	96.18	3.13	2.48
+5CNL(ours)	85.64	96.84	2.71	2.03
ResNet101	85.05	96.70	4.29	3.15
+5NL	85.53	96.65	5.03	3.96
+5CNL(ours)	86.73	96.75	4.61	3.52

Experimental results of ablation. Top1 and top5 accuracy (%) on CUB-200-2011.

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(a) CUB-200-2011

Method	Anno.	1-Stage	top1
Part-RCNN	✓	×	81.6
PN-CNN	✓	×	85.4
DVAN	×	×	79.0
B-CNN	×	×	84.1
PDFR	×	×	84.5
RACNN	×	×	85.3
RAM	×	×	86.0
DeepLAC	✓	✓	80.3
PA-CNN	✓	✓	82.8
SPDA-CNN	✓	✓	85.1
NAC	×	✓	81.0
FCAN	×	✓	84.7
MACNN	×	✓	86.5
MAMC	×	✓	86.5
Ours	×	✓	**86.7**

(b) Stanford Dogs

Method	Anno.	1-Stage	top1
PDFR	×	×	72.0
DVAN	×	×	81.5
RACNN	×	×	**87.3**
NAC	×	✓	68.6
FCAN	×	✓	84.2
MAMC	×	✓	85.2
Ours	×	✓	85.2

(c) Stanford Cars

Method	Anno.	1-Stage	top1
DVAN	×	×	87.1
B-CNN	×	×	91.3
RACNN	×	×	92.5
PA-CNN	✓	✓	92.8
FCAN	×	✓	89.1
MAMC	×	✓	93.0
Ours	×	✓	**93.1**

Ours: ResNet101+5CNL.
"Anno." stands for using extra annotation(bounding box or part) in training.
"1-Stage" indicates whether the training can be done in one stage.

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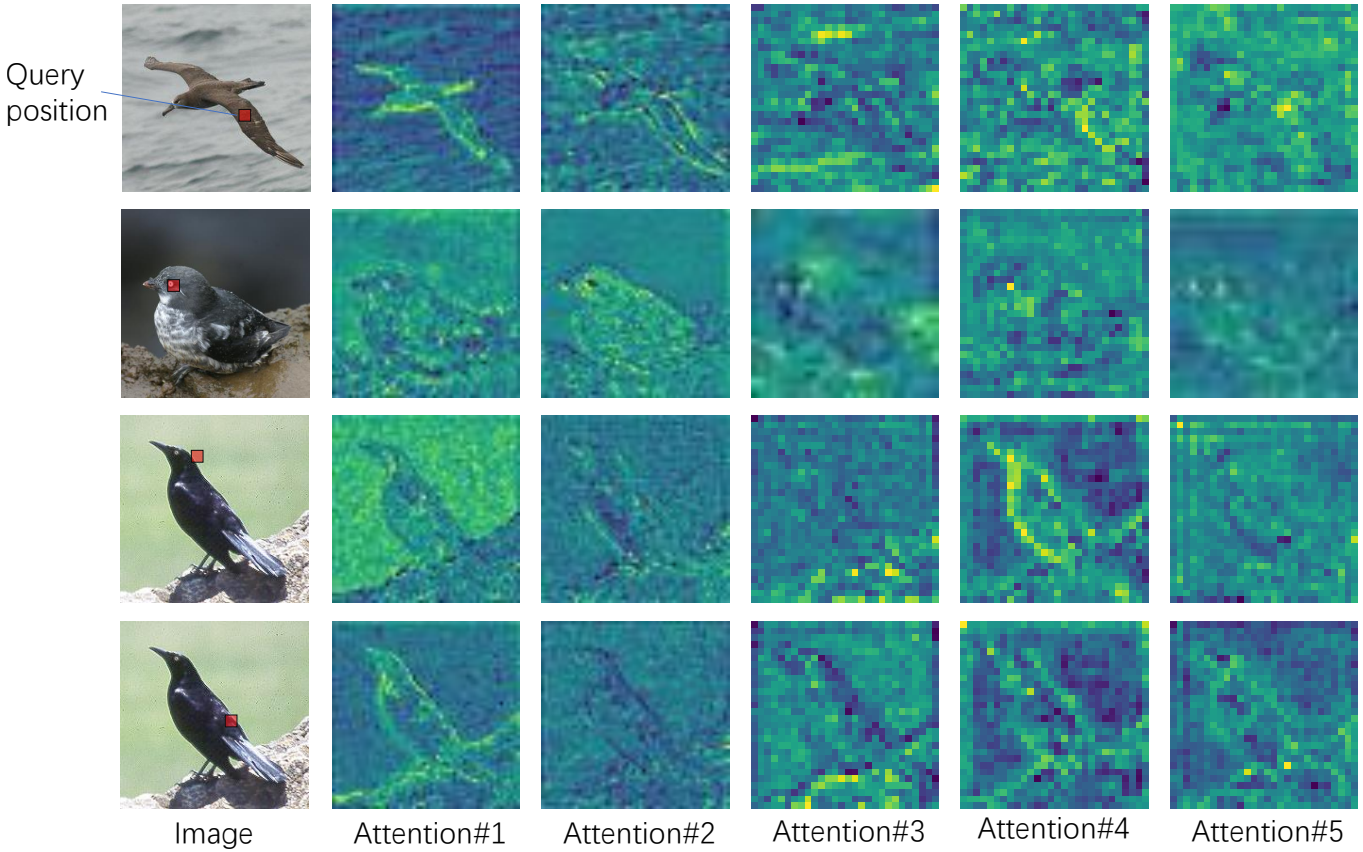
Visualization



Attention maps for different query positions (red blocks) in CUB.
Column 1: the input image.
Column 2 to 6: attention maps from shadow response layers to deep response layers.

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Conclusion I

(1) We have introduced a general formulation of Non-Local (NL).
(2) We have proposed a novel Cross-layer Non-Local (CNL) module for fined-grained image recognition.
(3) Our model could learn more discriminative features and adaptively focuses on multi-scale parts of images by associating cross-layer features.
(4) Our CNL eases the heavy association cost of original NL and other modules, but produces more competitive results and surpasses state-of-the-art results.
(5) Our codes can be found at *github.com/FouriYe/CNL-ICIP2020*.

Thanks!